

UNIVERSITAT DE BARCELONA

FUNDAMENTALS OF DATA SCIENCE MASTER'S THESIS

Algorithmic Fairness in ICU Mortality Prediction

A Multi-Model Analysis of Racial Disparities, Structural Bias, and Mitigation Strategies Using MIMIC-IV

Author:
Ino GRGUREVIĆ

Supervisor:
Dr. Laura IGUAL MUÑOZ

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Abstract

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by Ino GRGUREVIĆ

Machine learning models are increasingly used to predict in-hospital mortality for intensive care patients, which raises the question of whether they perform equally well for every population subgroup. This work uses MIMIC-IV a large and freely available critical care database to compare four model architectures. Two of these models, **XGBoost** and the **deep feedforward network**, use static summary features computed over the first twenty-four hours, while the **LSTM** and **Transformer** process the full hourly time series instead, and studies whether their predictions show systematic performance gaps across racial groups, what causes any such gaps, and whether they can be reduced. Across all four models the same small racial gap emerges, modest in size and consistent regardless of architecture. A series of experiments locates the source of this gap in the data itself rather than in any model, and shows that it responds only to changes at the level of data collection, not to adjustments made to the model afterward. The findings point to more equitable clinical measurement and data collection as the most reliable route to fairer ICU prediction.

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Chapter 1

Introduction

The adoption of electronic health records (EHR) over the past twenty years has produced a huge quantity of digital clinical data, and with it the opportunity to apply machine learning to a variety of medical prediction tasks. Among these, the prediction of in-hospital mortality for patients admitted to intensive care units is the focus of this work. An accurate early warning system can help clinicians prioritize the most critical patients, allocate limited resources more effectively, and intervene before a patient deteriorates. These systems are now being developed for hospitals to implement and use.

With this opportunity of machine learning models comes a serious responsibility. A prediction model that performs well on average may nonetheless perform poorly for a particular sub-population of patients. If a model is systematically less accurate for one racial group than for another, it risks reinforcing this bias within actual deployments. A model that fails to identify deteriorating patients in a protected group will provide less benefit to that group, even as it appears to perform well when measured across the whole population. For this reason, the fairness of clinical prediction models has become a central concern in the field of machine learning within health-care.

1.1 Research Questions

The thesis is built around two primary research questions:

1. Do ICU mortality prediction models exhibit systematic performance disparities, primarily across racial groups but also across gender and age groups, when trained on MIMIC-IV?
2. Does the choice of model architecture, in particular the contrast between static feature-based models and temporal sequence models, affect the magnitude of these disparities?

Several secondary questions arise naturally from the experimental work. The first concerns the role of missing data: Black and White patients are not monitored with equal frequency in the ICU, and this work investigates whether the resulting pattern of missing measurements is itself something the model learns to exploit when predicting mortality. The second concerns mitigation: if a disparity exists, can it be reduced by imputing the missing data, by balancing the training set, or by applying an explicit fairness constraint during training? The third concerns the deeper question of whether race is encoded in the clinical measurements themselves, such that a model can recover a patient's race even when it is never told what that race is.

1.2 The MIMIC-IV Dataset

All experiments in this thesis use MIMIC-IV version 3.1, the most recent release of the Medical Information Mart for Intensive Care database at the time of writing. MIMIC-IV is a large electronic health record dataset developed by the Massachusetts Institute of Technology Laboratory for Computational Physiology in collaboration with the Beth Israel Deaconess Medical Center in Boston. It contains de-identified records for patients admitted to the hospital and its critical care units, with detailed information on demographics, vital signs, laboratory results, medications, procedures, and clinical outcomes [1, 2].

The database is organized into modules. The hospital module contains data recorded across the whole hospital stay, including admission and discharge information, patient demographics, and laboratory measurements. The intensive care module contains the higher-frequency data collected during ICU stays, including the vital sign measurements. Vital signs are drawn from the `chartevents` table, which records every charted observation during a patient's time in the ICU.

MIMIC-IV is widely used precisely because it is both large and freely available, which makes it possible for researchers to reproduce and build upon published work. The same database has supported a substantial body of research on clinical prediction, including the benchmarking of mortality models on its predecessor, MIMIC-III [6]. Despite being open access, it is not unconditional: researchers must become credentialed PhysioNet users, complete a recognized human research protection course, and sign a data use agreement prohibiting re-identification or unauthorised sharing.

Chapter 2

Background and State of the Art

This chapter reviews the existing research that this thesis builds on, covering key examples of algorithmic bias in clinical machine learning, the sources of bias present in health data, and the benchmarking of recurrent models for ICU mortality prediction.

2.1 Algorithmic Fairness in Healthcare

A study by Obermeyer and colleagues examined a commercial algorithm used across the United States to identify patients with complex health needs who might benefit from additional care [3]. The authors found that the algorithm exhibited substantial racial bias: at any given risk score produced by the algorithm, Black patients were in fact considerably sicker than White patients, as measured by the number and severity of their underlying conditions. The cause was traced to the algorithm's use of healthcare spending as a proxy for healthcare needs. This study illustrates several themes that recur throughout the fairness literature, the most important being that bias can arise from technical choices that appear neutral, such as the selection of a prediction target.

Chen and colleagues provide a broader framework for considering ethics and fairness throughout the machine learning pipeline in healthcare [4]. They identify points at which fairness concerns arise at every stage, from the initial choice of which problem to solve, through the collection of data and the training of models, to the monitoring of systems after deployment. Importantly, they emphasise that fairness cannot be reduced to a single number. Different mathematical definitions of fairness can be mutually incompatible, so that satisfying one may require violating another, and the appropriate definition depends on the clinical context and on the values of those affected. They argue that addressing fairness requires attention to the social context in which a model is built and used, not just to the model itself.

Rajkomar and colleagues offer a complementary perspective from the principle of distributive justice [7]. They describe how the design of a model, biases present in the training data, and the way model predictions interact with clinicians and patients can each contribute to disparities. Rather than treating fairness as a constraint to be satisfied passively, they argue that machine learning should be used proactively to advance health equity, and they outline several technical notions of fairness, including equal performance across groups, meaning that a model should achieve similar accuracy for each subpopulation, and equal allocation of resources, meaning that the benefits or treatments triggered by a model's predictions should not disproportionately favor one group over another.

2.2 Sources of Bias in Clinical Data

Bias in clinical data may stem from data and algorithmic sources, including measurement bias, sampling bias, and modeling bias, as well as from structural factors arising from underlying inequities in healthcare delivery and data generation.

Sjoding and colleagues demonstrated that pulse oximeters, which are used as monitoring devices in both the general hospital and intensive care units to estimate blood oxygen saturation non-invasively, systematically overestimate oxygen saturation in patients with darker skin pigmentation [5]. This systematic measurement error has direct implications for the quality of the data and, consequently, for machine learning models trained on it. It is a central motivation for one of the simulation experiments presented later in this thesis.

2.3 Mortality Prediction and Benchmarking in MIMIC

Mortality prediction is one of the most commonly studied prediction tasks in the ICU setting, since it is clinically meaningful. Harutyunyan and colleagues established a widely used set of benchmark tasks on MIMIC-III, including in-hospital mortality prediction, and proposed strong baseline models built on recurrent neural networks [6]. Their work demonstrated that models operating on the full time series of vital signs, rather than on summary statistics alone, could achieve strong performance, and it helped to establish recurrent architectures as a standard approach for clinical time series.

Chapter 3

Methodology

This chapter describes the engineering of features, the four selected prediction models, the construction of the study cohort and the implementation.

3.1 Feature Engineering

Two representations of each patient’s first twenty-four hours in the ICU were constructed from the same five vital signs: a tabular representation for the non-sequential model and a time series representation for the sequence models. The five vital signs, heart rate, respiratory rate, oxygen saturation, body temperature, and mean arterial pressure, were extracted from the `chartevents` table. They were chosen because they are routinely measured in essentially all ICU patients, feature in established severity-of-illness scoring systems, and appear in MIMIC-IV with enough frequency to support analysis. Outlying values arising from charting errors or sensor defects were treated as missing by applying physiological bounds to each sign: heart rate from 20 to 300 beats per minute, respiratory rate from 4 to 60 breaths per minute, oxygen saturation from 50 to 100 percent, temperature from 25 to 45 degrees Celsius, and mean arterial pressure from 20 to 200 millimetres of mercury.

For the tabular representation, each patient was summarized by the mean, standard deviation, minimum, and maximum of each vital sign over the first twenty-four hours, which together with age at admission produced a twenty-one-dimensional feature vector. This representation discards temporal ordering but captures the overall level and variability of each sign. For the time series representation, each patient was represented by a matrix of twenty-four rows, one per hour, and ten columns. The first five columns held the average value of each vital sign during that hour. Missing values were filled by carrying the most recent earlier measurement forward in time. For the hours preceding a patient’s first recorded measurement of a given vital sign, where no earlier value exists to carry forward, that patient’s earliest available measurement was instead carried backwards to fill the leading gap. Only when a patient had no measurement of a given vital sign at all was the population median used. The remaining five columns held binary indicators recording, for each vital sign, whether a genuine measurement had been taken in that hour or whether the value had been imputed. This explicit encoding of missingness as separate channels allows the models to learn from the monitoring pattern, that is, which vital signs were actively measured, as well as from the measurement values themselves. The indicators are preserved even in experiments where missing values are imputed, since they retain information about the actual clinical monitoring rather than about the filled-in values [6]. Figure 3.1 shows how this monitoring pattern varies by demographic group.

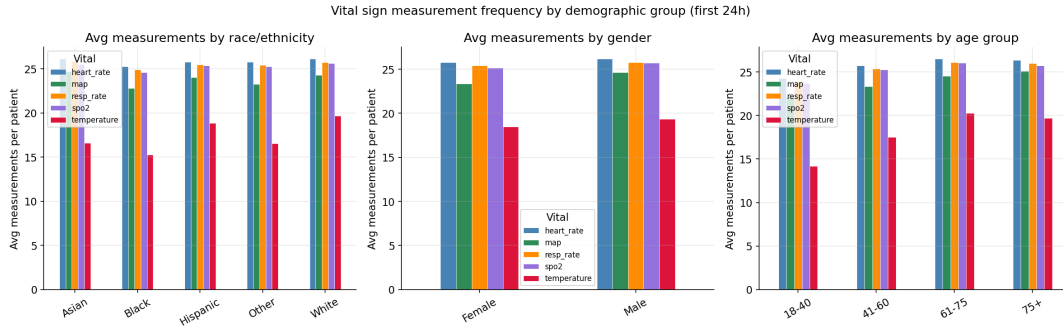


FIGURE 3.1: Vital sign measurement frequency by demographic group during the first twenty-four hours of ICU admission. Each bar represents the average number of measurements per patient for each vital sign, grouped by race and ethnicity (left), gender (center), and age group (right).

3.2 Models

Four architectures were evaluated under identical conditions, differing only in how they process the input and not in the data, the folds, or the evaluation.

XGBoost, an implementation of gradient-boosted decision trees widely used for tabular prediction, was trained on the twenty-one-dimensional feature vector, using three hundred trees, a maximum depth of four, a learning rate of 0.05, and row and column subsampling at 0.8 to reduce overfitting [8].

The **deep feedforward neural network** flattens the twenty-four by ten time series matrix into a single vector of 240 values and processes it through fully connected layers with batch normalization, rectified linear activations, dropout, and a residual connection. It ignores temporal structure and therefore serves as a point of comparison against the models that exploit it.

The **bidirectional Long Short-Term Memory network** has three layers and a hidden state of 128 units in each direction, processing the input as a sequence of twenty-four hourly steps and using both earlier and later context to form its representation of each hour, after which the final state passes through a small feedforward classifier [6].

The **Transformer encoder** treats each hour as a token. It projects each ten-dimensional vector to a model dimension of 128, adds a learnable positional embedding (one per hour), and applies three encoder layers, each with eight attention heads and an inner feedforward dimension of 512, using pre-layer-normalization for training stability. The hourly representations are then averaged across time and passed to a small feedforward head for classification [9]. Because attention is computed across all twenty-four hours simultaneously rather than sequentially, the model can relate any two time points directly, regardless of their distance.

3.3 Cohort Construction

The study cohort was derived from MIMIC-IV version 3.1. Only the first ICU stay of each patient was used, which prevents a single patient from appearing in both the training and the test data and so avoids leakage across the split. Patients younger than eighteen years at admission were excluded, because the physiology and clinical management of pediatric critical care differ substantially from those of adults, and

stays shorter than four hours were excluded, since these often reflect administrative or observational circumstances rather than genuine intensive care and provide too little data for meaningful prediction.

Race and ethnicity were taken from the hospital admission records and consolidated into five broad groups (White, Black, Hispanic, Asian, and “Other”), together with an Unknown category for patients whose race was not recorded. The Unknown category was excluded from all analyses, because the absence of recorded demographic information may itself reflect a bias in data collection and cannot be assigned to any group. The “Other” category was kept in the initial cohort but excluded from the four-way racial comparison that forms the core of the fairness analysis. After these steps the final four-way cohort comprised 52,852 patients: 42,625 White (81 percent), 5,990 Black (11 percent), 2,268 Hispanic (4 percent), and 1,969 Asian (4 percent), with an overall in-hospital mortality rate of 9.4 percent. This composition is itself relevant to the fairness analysis, since the substantial imbalance between the White majority and the much smaller minority groups has consequences both for the statistical power of any comparison between groups and for the results themselves.

3.4 Implementation

All experiments were implemented in Python. The gradient-boosted trees used the XGBoost library, and the three neural networks were implemented from scratch in PyTorch and trained on a server equipped with two RTX 3090 GPUs. Data preparation and preprocessing, including feature scaling, imputation, and cross-validation splitting, used the pandas, NumPy, and scikit-learn libraries, and statistical significance testing used the statsmodels library. All code for this thesis is publicly available at https://github.com/IGLucky/TFM_UB.

Chapter 4

Experimental Setup

4.1 Cross-Validation Protocol

All models were evaluated using five-fold stratified cross-validation, in which the data was divided into five parts and each part used in turn as a test set while the remaining four were used for training. The division was stratified on a combined label encoding both the mortality outcome and the racial group, ensuring that every fold contained a representative number of patients from each group and each outcome. A single random seed was used in all experiments, so that differences in performance cannot be attributed to differences in the splits.

4.2 Evaluation Metrics

The main measure of model performance is the area under the precision-recall curve (AUPRC). In a dataset with large class imbalance, this is generally more informative than the area under the receiver operating characteristic curve. The AUPRC focuses on the ability to identify rare positive cases without raising too many false alarms, which is the question that matters for an early warning system. The area under the receiver operating characteristic curve (AUROC) is also reported throughout as a secondary measure, since the two together give a fuller picture.

Three measures of group fairness were computed, each comparing Black and White patients at a fixed decision threshold, following the separation-based definitions of Liu and colleagues [10]. Writing TPR_g , FPR_g , and $\text{FNR}_g = 1 - \text{TPR}_g$ for the true positive, false positive, and false negative rates within group g , and taking the range across groups (which for the two groups here is simply the absolute difference), the three metrics are

$$\begin{aligned} \text{Equal opportunity} &= | \text{TPR}_b - \text{TPR}_w |, \\ \text{Equalized odds} &= \max(| \text{TPR}_b - \text{TPR}_w |, | \text{FPR}_b - \text{FPR}_w |), \\ \text{BER equality} &= | (\text{FPR}_b + \text{FNR}_b) - (\text{FPR}_w + \text{FNR}_w) |, \end{aligned}$$

where b and w denote the Black and White groups. **Equal opportunity** captures whether the model is equally good at catching the patients who die in each group, that is, whether the true positive rates match. **Equalized odds** is a stricter criterion that also requires the false alarm rates to match, taking whichever of the two differences is larger. **Balanced error rate (BER) equality** compares the combined error load, the sum of the false positive and false negative rates, between the groups. For all three, a smaller value is a better result, with zero indicating perfect parity.

4.3 Simulation Experiments

Beyond measuring performance and fairness, a series of experiments was designed to investigate why the disparities occur.

In the **retrained imputation experiments**, the LSTM was retrained from scratch on data in which the missing measurements of Black patients had been filled in. Two variants were tested. The first (Variant A) imputed measurements so that the monitoring frequency of Black patients matched that of White patients, representing a realistic scenario of equal monitoring. The second (Variant B) filled every remaining gap, representing an idealised scenario of complete data. In both variants the imputed values were drawn from the medians of Black patients, so that the simulated physiology was not altered and only the pattern of missingness changed.

In the **bias simulation scenarios**, three clinically motivated forms of bias were simulated and applied to Black patients. The first applied a systematic downward shift of one to five percentage points to the oxygen saturation values of Black patients, motivated directly by the documented bias of pulse oximeters in patients with darker skin pigmentation [5]. The second removed all temperature measurements for Black patients, and the third added random noise of increasing magnitude to their heart rate measurements. Each scenario was run twice: once on the frozen baseline model with weights held fixed, and once with the LSTM retrained from scratch under the biased data, in order to distinguish the immediate effect of a perturbation from the effect that the same bias would have if it were present throughout training.

Finally, two further experiments probed the structure of the disparity. In the **subsampling experiment**, the number of White patients in each training fold was randomly reduced to approximately six thousand, to match the number of Black patients, in order to test whether the performance gap persists when the two groups are equally represented in training. In the **race prediction experiment**, models were trained to predict whether a patient was Black using only the vital sign data, with no demographic information provided, and were run both with and without the missingness indicators in order to see whether any racial signal is encoded in the pattern of measurement, in the values of the measurements, or in both.

4.4 Mitigation through Adversarial Debiasing

As a strategy for reducing the disparity, an adversarial form of the LSTM was implemented, following the adversarial debiasing approach of Zhang and colleagues [11]. A second network, the adversary, is trained alongside the main model and attempts to predict the patient's racial group from the internal representation that the main model forms, while the main model is trained with a combined objective: to predict mortality as accurately as possible while preventing the adversary from recovering race from that representation. A single parameter, λ , controls the strength of the fairness penalty, and four values were tested ($\lambda = 0, 1, 3$, and 5), ranging from zero, which reproduces the unconstrained model, up to a value strong enough to suppress most of the racial signal.

Chapter 5

Results and Discussion

This chapter presents the experimental findings and their discussion. It first establishes the overall predictive performance of the four models, then looks at the disparities between demographic groups, investigates where they come from and evaluates the mitigation strategies.

5.1 Bias Analysis

5.1.1 Overall performance

All four architectures achieve comparable overall results on the mortality prediction task. As shown in Figure 5.1, the AUROC values are all close to 0.81 for every model, and the AUPRC values lie between 0.417 and 0.433 against a positive prevalence of only 0.09. No architecture dominates the others. The two temporal models, the LSTM and the Transformer, hold a small edge in AUPRC over the static XGBoost and the feedforward network, but the differences are small and the curves are closely overlapping across almost the entire range. The practical implication is that the choice of architecture is not, in itself, a major determinant of predictive quality on this task, which makes the models a fair basis for comparing fairness properties rather than just raw accuracy. A comparison of the out-of-fold AUPRC distributions supports this. The Transformer was excluded from this pairwise model comparison, since its predictions were produced outside the three-model cross-validation export, so the test covers only XGBoost, the deep network, and the LSTM. Among these three, the only pairwise difference that reaches significance is the small advantage of the LSTM over the deep feedforward network (mean difference 0.011, $p < 0.001$), whereas the differences between XGBoost and either neural model are not significant ($p = 0.46$ and $p = 0.11$).

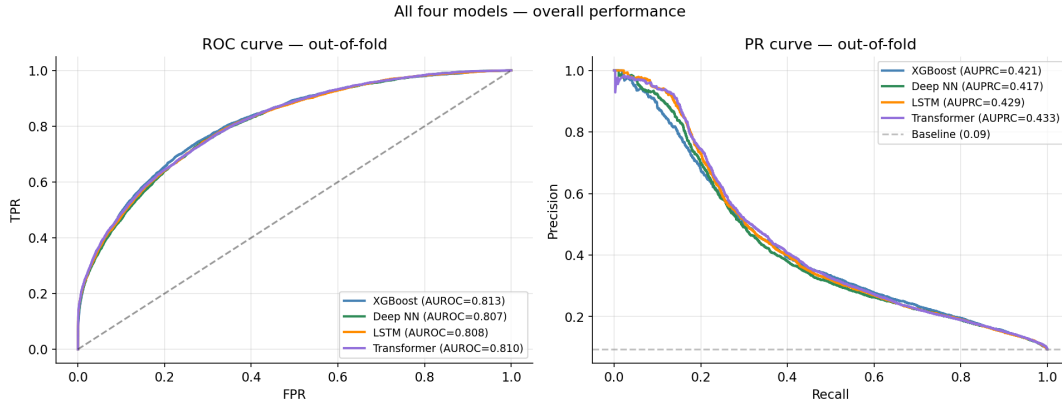


FIGURE 5.1: Overall out-of-fold ROC curves (left) and precision-recall curves (right) for all four models. The curves are tightly clustered, confirming that the architectures achieve very similar overall discrimination.

5.1.2 Disparities across racial groups

When performance is broken down by racial group, a consistent pattern is revealed, although it is modest in size. Figure 5.2 shows AUPRC by race for each model, and Figure 5.3 shows the same quantities with 95 percent bootstrap confidence intervals. Across all four architectures, White and Asian patients tend to receive the highest AUPRC, while Hispanic patients receive the lowest and Black patients sit in between. For the LSTM, for example, AUPRC is 0.435 for White patients, 0.441 for Asian patients, 0.403 for Black patients, and 0.395 for Hispanic patients. The size of this gap varies between models, from 0.021 for the Transformer to 0.033 for the LSTM (0.025 for XGBoost and 0.026 for the deep network), but in every model the gap runs the same way, with White patients scoring higher than Black patients.

Two points should be kept in mind. First, the confidence intervals for the smaller groups are wide, a direct consequence of the cohort imbalance described in Section 3.3: with fewer than 2,300 Hispanic and fewer than 2,000 Asian patients, the bootstrap intervals for these groups span a range comparable to the size of the gap itself, and they overlap substantially with the interval for White patients. Second, after correction for multiple comparisons across all pairwise groups, the Black-White difference did not reach statistical significance for any of the three cross-validated models, with FDR-corrected p -values of 0.344 for XGBoost, 0.450 for the deep network, and 0.562 for the LSTM. The Transformer was not included in this pairwise group test. The same pattern holds for every other pair of racial groups, none of which reached significance for any model. The full set of pairwise contrasts is reported in Figure A.1 (Appendix A). The disparity is best described as consistent in direction across architectures but small and not statistically substantial. In plain terms, every model leans the same way, but the differences are small and could be explained by chance, so this should be treated as a weak and consistent trend rather than as proof of unfairness.

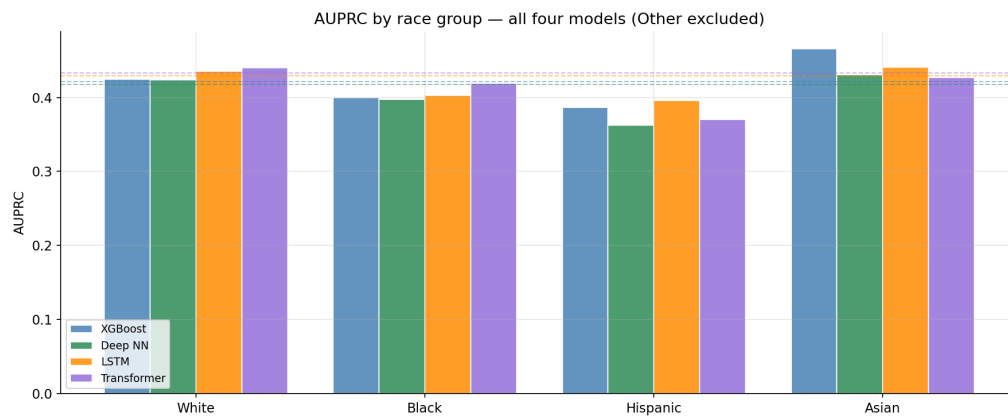


FIGURE 5.2: AUPRC by racial group for all four models (the “Other” group is excluded from the comparison). White and Asian patients tend to receive the highest scores and Hispanic patients the lowest, a pattern reproduced across architectures.

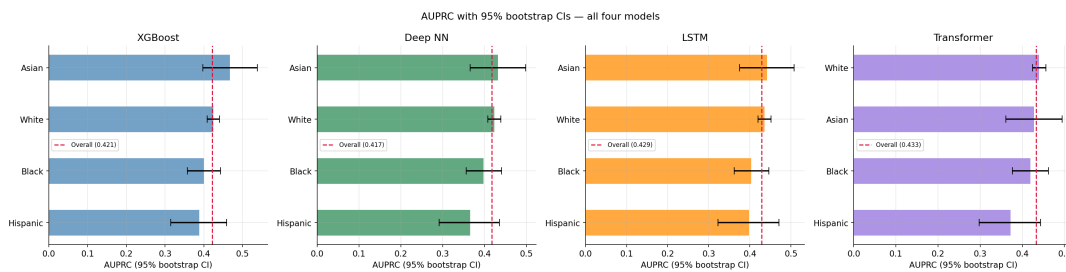


FIGURE 5.3: AUPRC by racial group with 95 percent bootstrap confidence intervals for each model. The intervals for the smaller groups are wide and overlap the overall mean, indicating that the disparities, while consistent in direction, are not statistically robust.

A further check asked whether this gap is a stable property of each group or partly an artifact of the metric, since AUPRC is sensitive to how common the positive class is. Evaluating the same models on the mirror-image task of predicting survival rather than death almost eliminates the gap (Appendix B).

5.1.3 Disparities across gender and age

Although the analysis centers on race, the same out-of-fold predictions allow a brief look at the other two demographic dimensions named in Research Question 1. The gender gap in AUPRC is very small, around 0.01 across the models (0.010 for XGBoost, 0.012 for the LSTM, and 0.015 for the deep network), with male patients scoring marginally higher than female patients. The age gap is the largest of the three demographic dimensions, around 0.05 (0.048 for the LSTM, 0.052 for the deep network, and 0.058 for XGBoost). Most of this age gap is a prevalence effect rather than a sign of unequal model quality: the in-hospital mortality rate rises sharply with age, from 4.2 percent in the 18 to 40 group to 13.8 percent in the 75 and over group, and because AUPRC depends on how common the outcome is, the youngest patients, who rarely die, are the hardest group to score highly on. The practical point is that race is not the largest demographic gap in this cohort.

5.1.4 Why does the disparity exist?

Having established that a modest and consistent disparity exists, the next set of experiments investigates its origin. They point towards the data and the way it is collected, rather than the models itself, as the source of the gap. The simulation and imputation experiments below are reported in their retrained form, where the imputed data is present throughout the training. The matching frozen-weight versions, where the change is applied only at test time, are in the Appendix C.

The first piece of evidence is that patients are not monitored equally. Figure 3.1 (Section 3.1) showed this briefly, but it is worth restating. Heart rate, respiratory rate, and oxygen saturation are charted for more than 99.6 percent of patient-hours in every racial group, so there is essentially no gap there. Mean arterial pressure is different: it is charted for 38.6 percent of White patient-hours but only 26.9 percent of Black patient-hours, a gap of nearly 12 percentage points, with the other groups in between. This one vital sign is therefore missing far more often for Black patients than for White patients, which means the missingness indicators in the model are not just a technical detail. They show a real difference in how closely each group is watched.

Another reason is that the data contains race information, which can be seen when trying to recover it directly. Figure 5.4 shows the result of training models to predict whether a patient is Black from the vital sign data alone, with no demographic information given. Both the LSTM and XGBoost recover race well above the chance level of 0.5. The LSTM reaches an AUROC of 0.687 with the missingness mask and 0.683 without it, and XGBoost scores similarly, at 0.702 with all features and 0.694 using only the mean of each vital sign. Two points stand out. First, removing the missingness indicators barely changes the result (the LSTM falls by only 0.004), so most of the racial signal is carried by the measured values themselves, not only by the pattern of which measurements were taken. Second, XGBoost recovers race almost as well from the means of the vital signs alone as from the full feature set, so the signal survives even in coarse summaries. A model may therefore behave differently across groups without ever being told a patient's race, because information correlated with race is already present in the measurements and their recording.

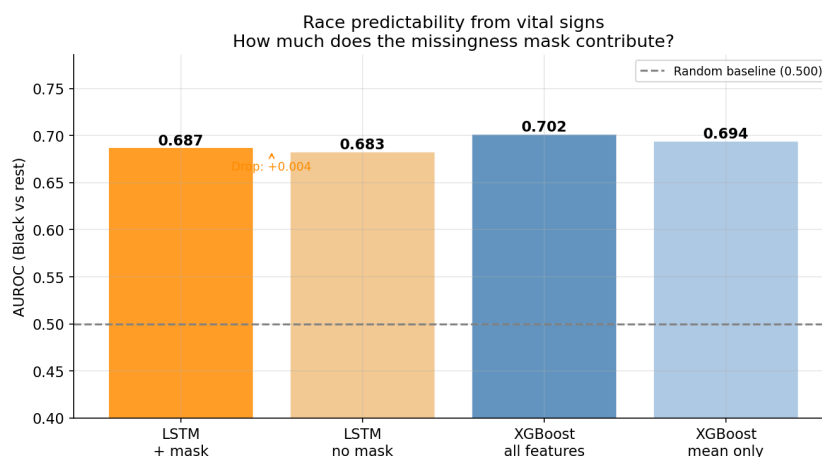


FIGURE 5.4: Predictability of race (Black versus the rest) from vital sign data alone. All configurations exceeded the chance baseline of 0.5, and removing the missingness mask produces only a marginal reduction, implying that the racial signal resides chiefly in the measurement values themselves.

5.2 Bias Mitigation

If the data carries group information, one simple explanation for the White advantage is that the models see far more White patients during training. Figure 5.5 tests this by retraining each model with the White training set reduced to roughly six thousand patients, matching the number of Black patients. Equalizing the sample size narrows the gap for three of the four models, most clearly for the LSTM, where it falls from 0.033 to 0.020 (the Transformer is the exception, moving slightly from 0.021 to 0.023). The gap does not close, though, and the narrowing comes mainly from a fall in White performance rather than any gain for Black patients. Representation imbalance therefore contributes to the advantage but is only a partial explanation, and cutting the majority group’s data is plainly not a route to a fairer model.

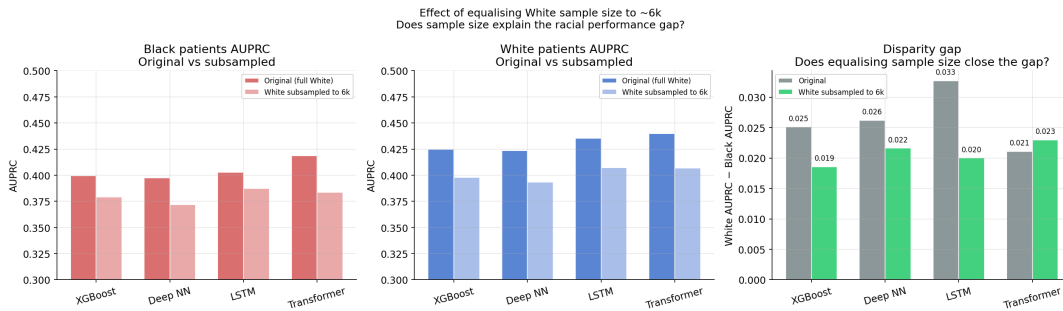


FIGURE 5.5: Effect of subsampling the White training set to match the size of the Black group. Black AUPRC (left) and White AUPRC (center) both fall under subsampling, and the disparity gap (right) narrows for most models, indicating that sample size is only a partial explanation for the disparity.

A more specific data problem is bias in the measured values themselves. Pulse oximeters are known to overestimate oxygen saturation in patients with darker skin pigmentation [5], a bias that Sjoding and colleagues found already present in historical records. Retraining the LSTM from scratch on data in which the oxygen saturation of Black patients has been shifted downward, so that the bias is present throughout training, has only a small effect: even a five-point shift changes Black AUPRC by about -0.012 . A bias that the model can learn around during training is therefore largely absorbed, at least in terms of predictive performance.

The picture is different when the same shift is applied only at test time to a frozen model, standing in for a bias that appears suddenly at deployment: there the degradation grows with the size of the shift and becomes significant, widening the White-Black gap from 0.033 to 0.069 (Appendix C). The contrast is the useful part: a measurement bias hurts the affected group most when it is new relative to the training data, as when a model is moved to another hospital or the monitoring equipment changes after deployment.

Removing temperature measurements and adding noise to heart-rate data did not really change the results after the model was retrained. The results using the original fixed model weights are shown in the Appendix C.

The next experiment turns from biased values back to missing ones and asks the most direct question: if the monitoring gap is the problem, does filling it in help? Figure 5.6 shows two retrained variants. Imputing Black patients’ missing measurements so that their monitoring frequency matches that of White patients (Variant A) leaves the disparity essentially untouched: the White-Black gap moves

only from 0.032 to 0.034 (this 0.032 baseline is from the retraining run and matches, to rounding, the 0.033 reported earlier), and Black AUPRC is indistinguishable from the baseline. Filling every remaining hourly gap with Black-patient medians (Variant B) is actively harmful: Black AUPRC collapses from 0.403 to 0.281. Because the missingness indicators are kept in both variants, the collapse is not from losing the monitoring signal; it comes from flattening long stretches of each series to a constant median, which replaces real within-stay variation with flat lines. The frozen-weight version of this full-imputation experiment is even more damaging and is reported in Appendix C. The lesson is that aggressively completing the data is worse than leaving it sparse, which argues against naive imputation as a fairness fix.

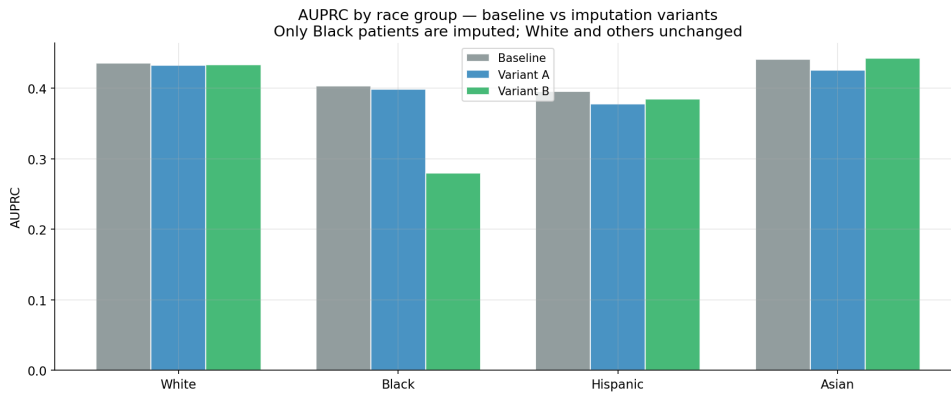


FIGURE 5.6: AUPRC by racial group after retraining the LSTM on imputed data. Only Black patients are imputed; the other groups are unchanged. Matching White monitoring frequency (Variant A) leaves performance almost unchanged, whereas filling every gap with median values (Variant B) sharply degrades Black performance.

Taken together, these experiments place the gap in the data and its collection rather than the model: group identity is recoverable from the measurements, unequal sample size explains only part of the advantage, a measurement-value bias matters most when it is new to the model, and directly repairing the missingness either changes nothing or makes things worse.

5.2.1 Adversarial debiasing

The principal mitigation strategy evaluated here is adversarial debiasing, in which the LSTM is penalized for encoding race in its internal representation. Figure 5.7 summarizes the effect of increasing the fairness penalty λ . The penalty does what it is designed to do: as λ rises, the adversary becomes steadily less able to recover race from the representation (bottom right), confirming that the constraint genuinely suppresses the racial signal rather than simply degrading the model. As λ rises, the overall AUPRC declines modestly, from 0.427 at $\lambda = 0$ to 0.400 at $\lambda = 5$ (top left), which is the expected price of the constraint. Two of the three group fairness metrics improve steadily over the same range (bottom left): the equalized odds gap falls from 0.117 to 0.075 and the balanced error rate gap from 0.035 to 0.017, while the equal opportunity gap, already small, shows no clear downward trend.

The Black-White AUPRC gap narrows in the same way, from 0.042 at $\lambda = 0$ to 0.010 at $\lambda = 5$ (top right), but the narrowing happens almost entirely because White AUPRC falls towards the Black level (from 0.436 to 0.404) while Black AUPRC barely moves (0.394 at both extremes). Adversarial debiasing, in other words, tends

to achieve parity by leveling down rather than by raising the disadvantaged group. Taking the trade-off as a whole, $\lambda = 3$ is a reasonable operating point, delivering a clear reduction in the fairness gaps while keeping overall AUPRC at 0.416, only marginally below the unconstrained value.

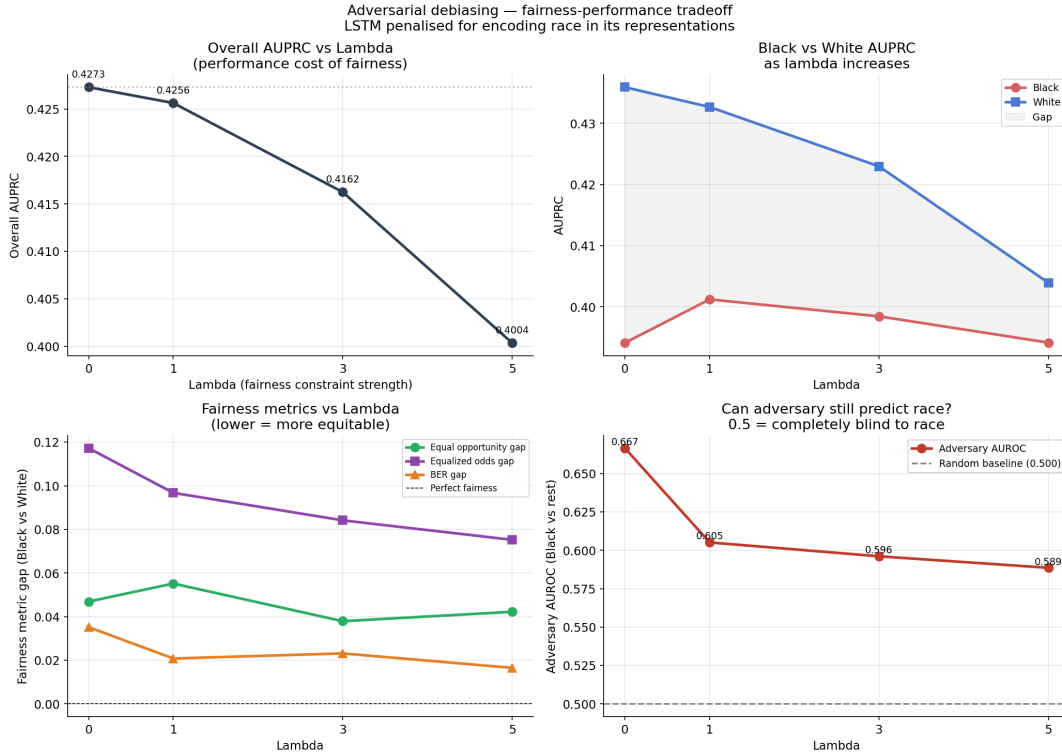


FIGURE 5.7: Adversarial debiasing of the LSTM. As the fairness penalty λ increases, overall AUPRC declines modestly (top left), the Black-White AUPRC gap narrows (top right), the fairness metric gaps shrink (bottom left), and the adversary’s ability to recover race falls towards chance (bottom right).

5.2.2 Comparison of mitigation strategies

This thesis tried four ways of reducing the racial gap, which fall into two families: changing the data (balancing the training set and imputing missing values) and constraining the model (adding race as an explicit feature, and adversarial debiasing). They differ mainly in how they fall short, because none of them raises the performance of Black patients in absolute terms.

The two data-side strategies do not help. Balancing the training set by subsampling White patients narrows the LSTM gap from 0.033 to 0.020 (Figure 5.5), but only by dragging White performance down to the Black level, not by improving Black performance. Imputation is worse: matching Black patients’ monitoring frequency to that of White patients leaves the gap essentially unchanged (Variant A), while filling in every missing value collapses Black AUPRC from 0.403 to 0.281 (Variant B, Figure 5.6). Repairing the data at the point where it is most unequal therefore does nothing useful and can do real harm.

The two model-side strategies are compared directly in Figure 5.8, which places the baseline LSTM, a model given race as an explicit feature, and the adversarially debiased model at $\lambda = 3$ side by side on the three fairness metrics. Adding race as a

feature is unreliable: it reduces the equalized odds gap (0.118 to 0.073) and the balanced error rate gap (0.031 to 0.025) but worsens the equal opportunity gap (0.056 to 0.073), improving one criterion while degrading another, exactly the incompatibility that Chen and colleagues warn against [4]. Adversarial debiasing at $\lambda = 3$ is the most consistent of the four strategies, achieving the smallest equal opportunity gap (0.038) and the smallest balanced error rate gap (0.023), and at only a modest cost in overall accuracy, although the equalized odds gap is lower for the race-as-feature model. Even so, as the previous section showed, it reaches this parity by leveling White performance down rather than by raising Black performance.

The common thread across all four is that the gap can be narrowed but never closed in the right direction: it shrinks only by lowering the White group.

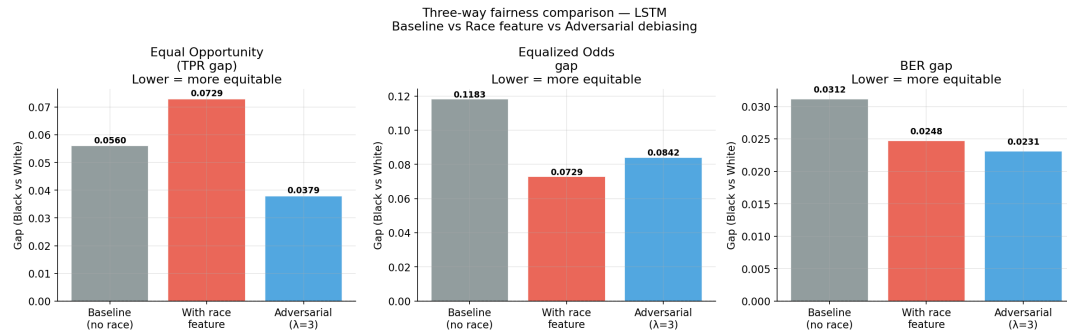


FIGURE 5.8: Three-way fairness comparison for the LSTM across the baseline (no race), the model with race as an explicit feature, and adversarial debiasing at $\lambda = 3$. Adversarial debiasing yields the smallest equal opportunity and balanced error rate gaps, whereas adding race as a feature worsens the equal opportunity gap.

Chapter 6

Conclusion

This work set out to find whether machine learning models that predict ICU mortality work equally well for different patient groups, and if not, why. Using MIMIC-IV and four models (XGBoost, a deep feedforward network, an LSTM, and a Transformer), it compared performance across racial, gender, and age groups, and then ran a series of experiments to understand where any differences come from and whether they can be reduced.

The first research question asked whether the models show systematic differences across groups. The answer is a yes. In every model, White and Asian patients received the highest AUPRC, Hispanic patients the lowest, and Black patients sat in between, with a White-Black gap of roughly 0.02 to 0.03. The direction of this gap was the same across all four models, but it was small, and after correcting for multiple comparisons it did not reach statistical significance for any model. The confidence intervals were especially wide for the smaller groups. The fair summary is that there is a weak and consistent trend rather than clear proof of unfairness.

The second research question asked whether the choice of model matters. It largely does not. The four architectures performed very similarly overall, and the size of the disparity did not depend in any clear way on whether the model used the full time series (LSTM, Transformer) or only summary features (XGBoost, deep network). Choosing a more complex model is therefore not a route to a fairer one.

The secondary questions concerned the data itself. The models could work out whether a patient was Black from the vital signs alone, above chance and almost as well without the missingness indicators as with them, which shows that signals linked to race sit in the measured values, not only in which measurements were taken.

On mitigation, none of the fixes worked well. Balancing the training set by reducing the number of White patients narrowed the gap a little, but mostly by making the model worse for White patients rather than better for Black ones. Filling in the missing measurements did nothing useful and, when taken to the extreme, was actively harmful. Adding race as an explicit feature was unreliable, improving one fairness measure while worsening another. Adversarial debiasing reduced the fairness gaps most consistently, but again it did this mainly by pulling the better-served group down rather than lifting the worse-served group up, and at some cost to overall accuracy.

Taken together, the results suggest that the differences seen here begin where the data is created, in how often and how accurately patients are monitored, rather than in the model that is trained afterwards. A simulated pulse oximeter bias widened the gap when it was new to the model, which supports this reading. The clearest message of the thesis is therefore that the most reliable way to make these models fairer is to improve the fairness of clinical measurement and data collection upstream, rather than to rely only on adjusting the model after the fact.

6.1 Limitations

Several limitations should be kept in mind when reading these conclusions. The mitigation, imputation, and simulation experiments were each run only once, not repeated, since each scenario required a full retraining of one or more models and repeating every scenario across multiple seeds was not feasible within the time available for this thesis. The model hyperparameters were not heavily tuned, so a bigger or better-tuned model might perform differently overall, though this would likely not change the pattern across racial groups. The bias simulations add an artificial change to the data to study its effect and they do not prove that this kind of bias is actually present in MIMIC-IV. Finally, the data comes from only one hospital in the United States, and grouping race and ethnicity into broad categories hides real differences within each group.

6.2 Future Work

Future work could extend this analysis in directions beyond the scope of this work. One direction is to test interventions at the point of data collection itself, such as actively equalizing monitoring frequency across groups, rather than imputing the gap afterward. A second is to apply a formal causal framework, modeling race's pathways through measurement frequency and measurement bias, to separate direct from indirect contributions to the gap. A third is external validation on a second hospital system, since MIMIC-IV reflects one academic center in one country.

Another direction concerns the model side: adversarial debiasing was only evaluated for the LSTM here, and extending it, or testing other mitigation techniques, across XGBoost, the deep network, and the Transformer would show whether the leveling-down pattern observed here is specific to one architecture or general to all four.

Bibliography

- [1] Johnson, A. E. W., Bulgarelli, L., Shen, L., Gayles, A., Shammout, A., Horng, S., Pollard, T. J., Hao, S., Moody, B., Gow, B., Lehman, L. H., Celi, L. A., & Mark, R. G. (2023). MIMIC-IV, a freely accessible electronic health record dataset. *Scientific Data*, 10(1), 1. <https://doi.org/10.1038/s41597-022-01899-x>
- [2] Johnson, A., Bulgarelli, L., Pollard, T., Gow, B., Moody, B., Horng, S., Celi, L. A., & Mark, R. (2024). MIMIC-IV (version 3.1). *PhysioNet*. <https://doi.org/10.13026/kpb9-mt58>
- [3] Obermeyer, Z., Powers, B., Vogeli, C., & Mullainathan, S. (2019). Dissecting racial bias in an algorithm used to manage the health of populations. *Science*, 366(6464), 447–453. <https://doi.org/10.1126/science.aax2342>
- [4] Chen, I. Y., Pierson, E., Rose, S., Joshi, S., Ferryman, K., & Ghassemi, M. (2021). Ethical machine learning in healthcare. *Annual Review of Biomedical Data Science*, 4, 123–144. <https://doi.org/10.1146/annurev-biodatasci-092820-114757>
- [5] Sjoding, M. W., Dickson, R. P., Iwashyna, T. J., Gay, S. E., & Valley, T. S. (2020). Racial bias in pulse oximetry measurement. *New England Journal of Medicine*, 383(25), 2477–2478. <https://doi.org/10.1056/NEJMc2029240>
- [6] Harutyunyan, H., Khachatrian, H., Kale, D. C., Ver Steeg, G., & Galstyan, A. (2019). Multitask learning and benchmarking with clinical time series data. *Scientific Data*, 6(1), 96. <https://doi.org/10.1038/s41597-019-0103-9>
- [7] Rajkomar, A., Hardt, M., Howell, M. D., Corrado, G., & Chin, M. H. (2018). Ensuring fairness in machine learning to advance health equity. *Annals of Internal Medicine*, 169(12), 866–872. <https://doi.org/10.7326/M18-1990>
- [8] Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 785–794. <https://arxiv.org/abs/1603.02754>
- [9] Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, L., & Polosukhin, I. (2017). Attention is all you need. *Advances in Neural Information Processing Systems*, 30. <https://arxiv.org/abs/1706.03762>
- [10] Liu, M., Ning, Y., Ke, Y., Shang, Y., Chakraborty, B., Ong, M. E. H., Vaughan, R., & Liu, N. (2024). FAIM: Fairness-aware interpretable modeling for trustworthy machine learning in healthcare. *Patterns*, 5(10), 101059. <https://doi.org/10.1016/j.patter.2024.101059>
- [11] Zhang, B. H., Lemoine, B., & Mitchell, M. (2018). Mitigating unwanted biases with adversarial learning. *Proceedings of the 2018 AAAI/ACM Conference on AI, Ethics, and Society*, 335–340. <https://arxiv.org/abs/1801.07593>

Appendix A

Pairwise Racial Group Comparisons

This appendix reports the full set of pairwise statistical comparisons between racial groups. Each pairwise AUPRC difference was tested with a permutation test, and p -values were corrected for multiple comparisons using the false discovery rate (FDR) procedure.

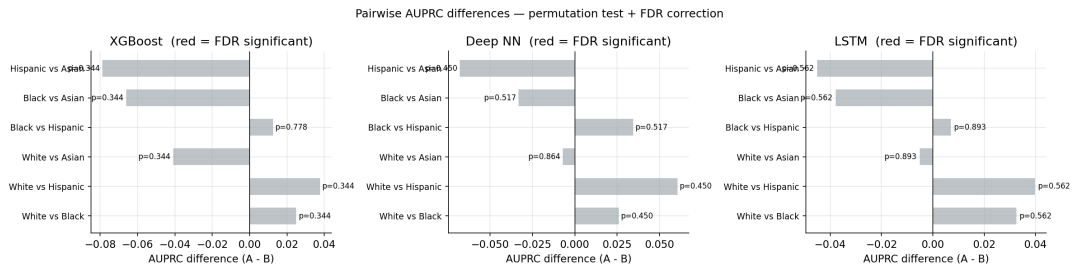


FIGURE A.1: Pairwise AUPRC differences between all racial group pairs, with permutation test p -values after FDR correction, for the three cross-validated models. No comparison reaches significance for any model.

Appendix B

Survival versus Mortality Prediction

A natural question is whether the measured gap reflects genuinely worse prediction for Black patients or whether it is partly an artifact of the metric. AUPRC depends on how common the positive class is, and the mortality rate differs slightly across groups. To test this, the identical models were evaluated on the mirror-image task of predicting survival rather than death, so that the positive class becomes the common outcome (a prevalence of about 0.91 rather than 0.09) while the data, the models, and the patients are otherwise unchanged. Figure B.1 shows AUPRC by racial group for both tasks.

Figure B.2 compares the White-Black AUPRC gap across the two tasks: for mortality the gap is 0.025 for XGBoost, 0.026 for the deep network, and 0.033 for the LSTM, whereas for survival the same gaps shrink to -0.002 , 0.001, and 0.003, that is, essentially zero and in one case marginally reversed. Even so, the shrinkage reflects a property of the metric, not evidence that the underlying disparity disappears. The gap is real, as the other experiments in this chapter show, but AUPRC's sensitivity to prevalence means its size should not be taken at face value

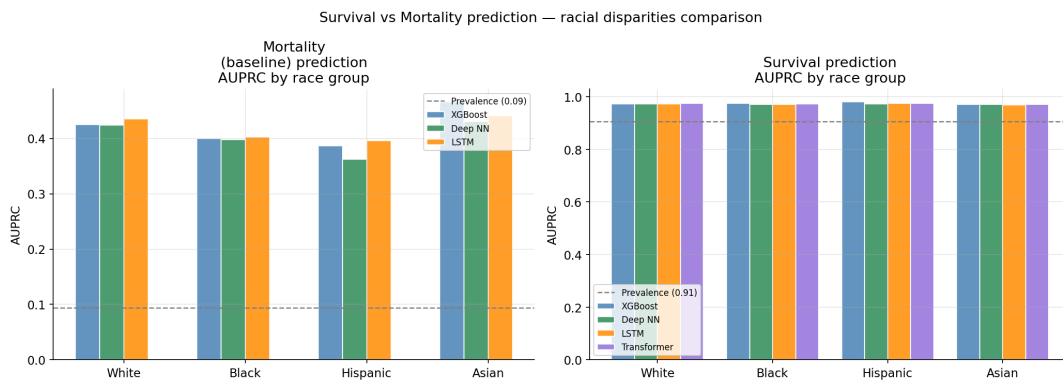


FIGURE B.1: AUPRC by racial group for mortality prediction (left) and the mirror-image survival prediction task (right). The dashed lines mark the positive-class prevalence in each task (0.09 for mortality, 0.91 for survival). The visible gaps between groups on the mortality task are almost absent when the common outcome is predicted instead.

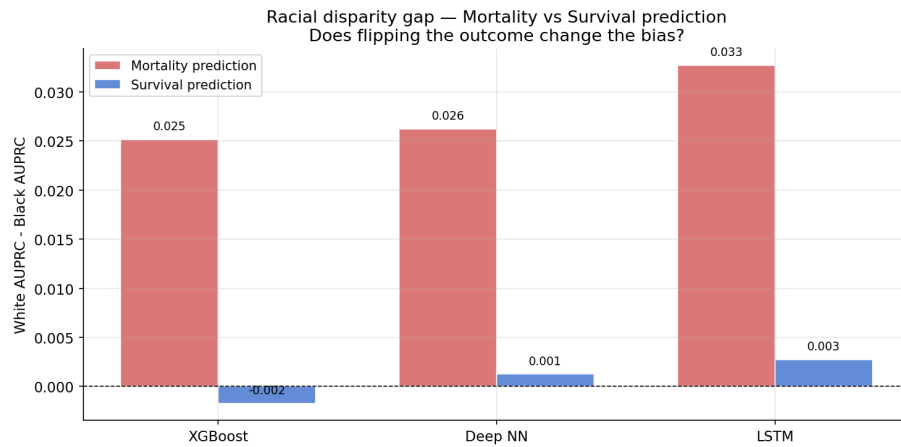


FIGURE B.2: White-Black AUPRC gap for mortality prediction compared with survival prediction. The gap of roughly 0.03 under mortality prediction collapses to near zero, and slightly below zero for XGBoost, once the outcome is flipped, indicating that much of the gap reflects the prevalence sensitivity of AUPRC rather than unequal discrimination.

Appendix C

Frozen-weight experiments

The simulation and imputation experiments in Section 5.1 are reported in the main text in their retrained form, where the altered data is present throughout training. For completeness, this appendix reports the frozen-weight versions, in which the change is applied only at test time to a model whose weights are held fixed. Because the model cannot adapt to the altered data, the frozen versions generally show a larger effect than their retrained counterparts. Figure C.1 summarizes all of them on a single axis, and the figures that follow give the detail for each scenario.

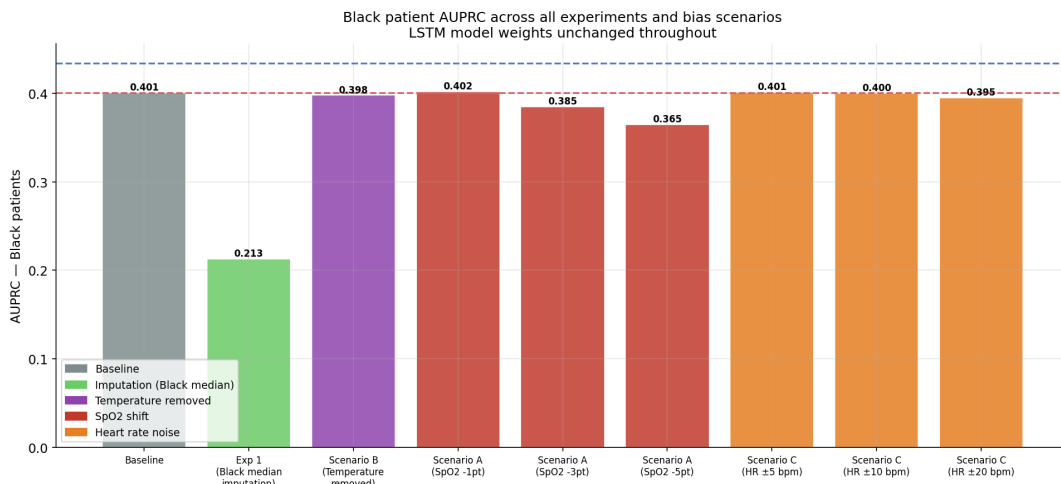


FIGURE C.1: Black-patient AUPRC across every frozen-weight experiment, with the LSTM weights held fixed throughout. The red line marks the Black baseline (0.401) and the blue line the White baseline (0.434). Only the full Black-median imputation (0.213) and the larger oxygen-saturation shifts move Black performance substantially; the temperature and heart-rate perturbations leave it close to baseline.

The oxygen-saturation shift (Scenario A) is the perturbation with the clearest dose-response.

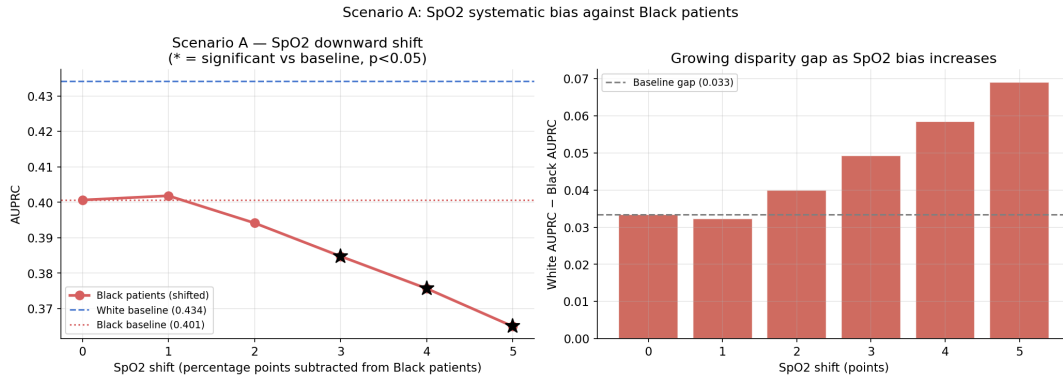


FIGURE C.2: Simulated systematic downward shift in the oxygen saturation of Black patients, applied at test time to the frozen baseline model. The decline in Black AUPRC (left) grows with the size of the shift and is significant from a three-point shift onward ($p = 0.02$ at three points, $p < 0.001$ at five points), and the disparity gap (right) widens from 0.033 to 0.069.

Removing temperature measurements (Scenario B) has almost no effect, which is expected given how rarely temperature is charted.

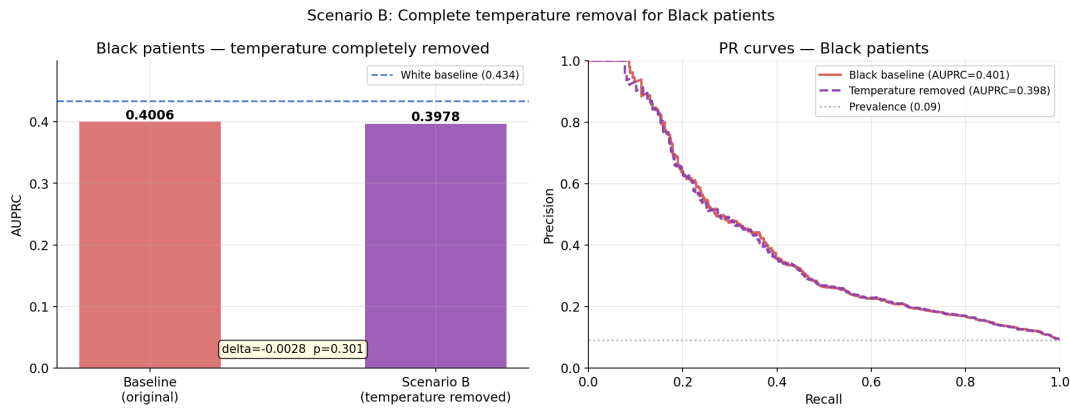


FIGURE C.3: Complete removal of temperature measurements for Black patients, frozen model. Black AUPRC falls only slightly, from 0.401 to 0.398 (delta -0.003 , $p = 0.30$), a non-significant change, consistent with temperature being charted for fewer than nine percent of patient-hours.

Adding heart-rate noise (Scenario C) matters only at large magnitudes.

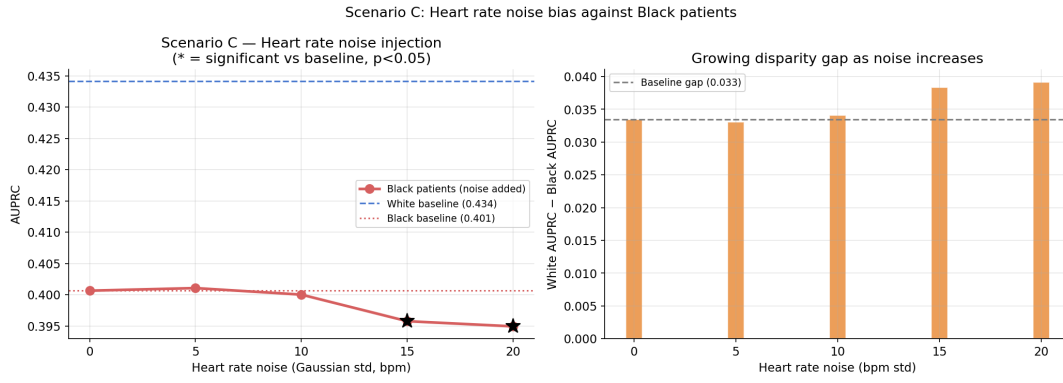


FIGURE C.4: Gaussian noise added to the heart-rate measurements of Black patients, frozen model. Black AUPRC (left) is unchanged at small noise levels and falls slightly at larger ones, becoming significant at 15 and 20 bpm (marked with stars; $p = 0.003$ and $p = 0.005$), while the disparity gap (right) grows only marginally above its baseline of 0.033.

The imputation experiments are the most damaging of the frozen scenarios. Filling Black patients' missing slots with the Black median collapses their AUPRC, and the mirror experiment of stripping White patients' measurements down to the sparser Black monitoring rate pulls White performance down to the Black level, which is a direct sign that monitoring frequency itself drives much of the gap.

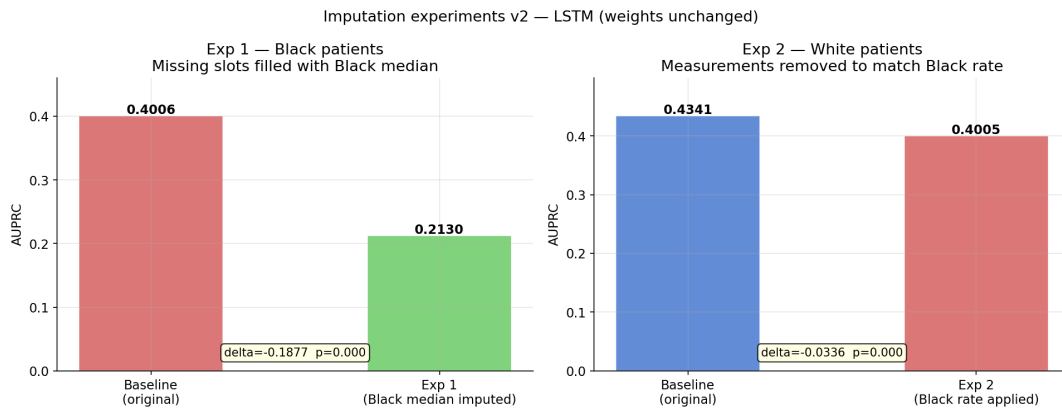


FIGURE C.5: Frozen-weight imputation experiments. Left: filling Black patients' missing slots with the Black median collapses their AUPRC from 0.401 to 0.213 (delta -0.188 , $p < 0.001$). Right: removing White patients' measurements to match the sparser Black monitoring rate drops White AUPRC from 0.434 to 0.400 (delta -0.034 , $p < 0.001$), bringing it down to the Black level.

Appendix D

Code Availability

All code used to produce the results in this thesis, including data extraction, preprocessing, model implementations, and the experiments reported in the main text and appendices, is publicly available at https://github.com/IGLucky/TFM_UB.